

Database Technologies – Assignment 8

Use sub-query to solve following problems.

Note: To solve below queries, use “sales” database

- Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

SELECT * FROM ORDERS WHERE CNUM=(SELECT CNUM FROM CUSTOMERS WHERE CNAME='Cisneros');

```
W2_99546_Pratiksha>USE SALES_DB;
Database changed
W2_99546_Pratiksha>SELECT * FROM ORDERS WHERE CNUM=(SELECT CNUM FROM CUSTOMERS WHERE CNAME='Cisneros');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

- Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

SELECT * FROM SALESPeOPLE WHERE CITY<>ANY(SELECT CITY FROM CUSTOMERS)LIMIT 2,3;

```
W2_99546_Pratiksha>SELECT * FROM SALESPeOPLE WHERE CITY<>ANY(SELECT CITY FROM CUSTOMERS)LIMIT 2,3;
+-----+-----+-----+-----+
| snum | sname | city   | comm |
+-----+-----+-----+-----+
| 1004 | Motika | London | 0.11 |
| 1007 | Rifkin | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

- Extract all the orders of Motika.

SELECT * FROM ORDERS WHERE SNUM=(SELECT SNUM FROM SALESPeOPLE WHERE SNAME = 'Motika');

```
W2_99546_Pratiksha>SELECT * FROM ORDERS WHERE SNUM=(SELECT SNUM FROM SALESPeOPLE WHERE SNAME = 'Motika');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

- Find all the orders of salespeople servicing customers in London.

SELECT * FROM ORDERS WHERE SNUM IN(SELECT SNUM FROM CUSTOMERS WHERE CITY='LONDON');

```
W2_99546_Pratiksha>SELECT * FROM ORDERS WHERE SNUM IN(SELECT SNUM FROM CUSTOMERS WHERE CITY='LONDON');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

- Find names and numbers of all salesperson who have more than one customer.

SELECT S.SNAME,S.SNUM FROM SALESPeOPLE S WHERE S.SNUM IN (SELECT SNUM FROM CUSTOMERS GROUP BY SNUM HAVING COUNT(*)>1);

```
W2_99546_Pratiksha>SELECT S.SNAME,S.SNUM FROM SALESPeOPLE S WHERE S.SNUM IN (SELECT SNUM FROM CUSTOMERS GROUP BY SNUM HAVING COUNT(*)>1);
+-----+-----+
| SNAME | SNUM |
+-----+-----+
| Peel  | 1001 |
| Serres | 1002 |
+-----+-----+
2 rows in set (0.00 sec)
```

- Select customers who have a greater rating than any other customer in Rome.

SELECT * FROM CUSTOMERS WHERE RATING>ANY (SELECT RATING FROM CUSTOMERS WHERE CITY='Rome');

```
W2_99546_Pratiksha>SELECT * FROM CUSTOMERS WHERE RATING>ANY (SELECT RATING FROM CUSTOMERS WHERE CITY='Rome');
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

7. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

SELECT * FROM ORDERS WHERE AMT>ANY (SELECT AMT FROM ORDERS WHERE ODATE='1990-10-04');

```
W2_99546_Pratiksha>SELECT * FROM ORDERS WHERE AMT>ANY (SELECT AMT FROM ORDERS WHERE ODATE='1990-10-04');
+-----+-----+-----+-----+-----+
| onum | amt | odate | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

8. Select those customers whose rating are higher than every customer in London.

SELECT * FROM CUSTOMERS WHERE RATING>ANY(SELECT RATING FROM CUSTOMERS WHERE CITY='LONDON');

```
W2_99546_Pratiksha>SELECT * FROM CUSTOMERS WHERE RATING>ANY(SELECT RATING FROM CUSTOMERS WHERE CITY='LONDON');
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```